Component C, Week 2, Task 1

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What the task asked

Run our existing code on `reference_results.csv` and `reference_truth.csv` (6 classifiers, 4 sample sizes, 100 replicates each, 12,000 rows) and report the four quantities at each sample size.

Results

n_sub	Winner-correct	Mean Kendall's tau	Mean optimism	Power (gb vs svm)	Sign-error rate (gb vs svm)
50	0.13	0.0177	-0.0352	0.03	0.05
100	0.10	0.2008	-0.0439	0.02	0.06
200	0.05	0.3358	-0.0216	0.00	0.11
500	0.08	0.4728	-0.0091	0.00	0.20

What each column means:

- **Winner-correct:** in how many of the 100 replicates the classifier with the highest average AUC is the one the truth file calls best. A tie for first place counts as not correct (this happened in 4, 4, 2 and 1 replicates).
- **Kendall's tau:** how closely the order of the six classifiers in a replicate matches their true order. 1 means the same order, 0 means unrelated, -1 means reversed. Averaged over the 100 replicates.
- **Optimism:** the winner's average AUC minus that same classifier's true AUC, averaged over the 100 replicates. Positive means the winner looks better than it really is.
- **Power and sign-error:** for gb vs svm, a paired t-test on each replicate's five fold scores. Power is how often it significantly favours gb (the truly better one). Sign-error is how often it significantly favours svm.

Which truth we used

Our existing code compares against each classifier's abundant-data (asymptotic) AUC, so Task 1 uses the asymptotic rows of the truth file. Optimism at n = 50 comes out at -0.0352, matching the -0.035 quoted in the Week 2 instructions for this version. gb and svm are the two classifiers with the highest asymptotic AUC (0.8368 and 0.8163), the same way the power pair was the top two classifiers in Week 1.

Worked example: replicate 1 at n = 50

Classifier	Five fold AUCs	Mean	True AUC (asymptotic)
dtree	0.375, 0.75, 0.4375, 0.6875, 0.625	0.5750	0.7378
gb	0.0, 0.1875, 0.4375, 0.6875, 0.875	0.4375	0.8368
knn	0.5625, 0.84375, 0.8125, 0.4375, 0.625	0.6563	0.7376
logreg	0.6875, 0.8125, 0.625, 0.625, 0.9375	0.7375	0.7886
rf	0.5, 0.75, 0.4375, 0.59375, 0.625	0.5813	0.8154
svm	0.625, 1.0, 0.75, 0.625, 0.625	0.7250	0.8163

- **Winner:** logreg (highest mean, 0.7375). The true best is gb, so this replicate counts as not correct.
- **Optimism:** 0.7375 - 0.7886 = -0.0511.
- **Kendall's tau:** of the 15 pairs of classifiers, 6 are in the same order as the truth and 9 are not, so tau = (6 - 9) / 15 = -0.2.
- **Power (gb vs svm):** fold differences are -0.625, -0.8125, -0.3125, 0.0625, 0.25. Their mean is -0.2875 and the standard error is 0.2002, so t = -1.436 with 4 degrees of freedom, p = 0.224. Not significant, so this replicate is no detection.

These are the values on this replicate's line in `Task1_per_replicate.csv`. The table above is the average of those 100 lines for each sample size.

What changed in the code

- The truth file now has an `n_sub` column that holds numbers and the word `asymptotic`, so it is read as text and stored by sample size.
- Six classifiers instead of three. The list comes from the truth file, nothing is typed in.
- The power pair is the top two classifiers by true AUC, with the better one first, so correct detections and sign errors are labelled the right way round.

Checks

- All four quantities were recomputed with a separate script and match exactly.
- The Week 1 edge-case tests still pass (15 passed, 0 failed).
- The reference file has no missing AUCs, duplicate rows or odd fold numbers, and every replicate has all 6 classifiers with 5 folds.

What the numbers show

Winner-correct is only 0.05 to 0.13 and does not improve with sample size (random guessing among six would give about 0.17). Optimism is negative at every size, and the sign-error rate rises to 0.20 at n = 500. The reason is that gb is the abundant-data winner but is not the best model at these sample sizes: logreg is best at 50 and 100 rows and svm at 200 and 500. A study that correctly picks logreg at 50 rows is marked wrong, and small-sample models score below their abundant-data values, so optimism goes negative. At n = 500 svm really does beat gb (0.786 against 0.757), so a test favouring svm is counted as a sign error only because the pair was labelled using abundant-data truth. Task 2 looks at this problem.